

TRISTAN ROMAN

DATA SCIENTIST | PROGRAM & FINANCIAL ANALYTICS | PRODUCTION DECISION-SUPPORT TOOLS

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PROFESSIONAL SUMMARY

Data scientist with a Master of Data Science (UC San Diego) and four years of progressive Northrop Grumman experience across program control, pricing, and contracts on DoD programs — hands-on domain expertise in program management, earned value management, and financial forecasting. Proficient in Python, SQL, and Git, with applied experience in statistical modeling, machine learning, ETL/ELT pipeline development, and interactive Streamlit and Tableau decision-support tools. Proven record of translating ambiguous business problems into rigorous analytical frameworks, critically evaluating data quality and assumptions, delivering actionable recommendations to technical and non-technical stakeholders, and taking ownership of methodology and outcomes. Currently extending an end-to-end analytics platform with AWS, Databricks (PySpark), Docker, and MLOps practices. Active security clearance.

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, scikit-learn), SQL, Git; JavaScript, Java, C#, HTML/CSS

Statistics & Machine Learning: Regression, classification, anomaly detection, time-series analysis and forecasting, hypothesis testing, reliability analysis, model evaluation and validation, recommender systems (SVD, SVD++, Bayesian Personalized Ranking), deep learning and reinforcement learning concepts

Data Engineering: ETL/ELT pipeline design, PostgreSQL, relational and star-schema modeling, schema validation, data quality and lineage, idempotent loads, reproducible workflows, CSV/JSON/fixed-width .DAT parsing, batch and real-time analytics concepts

Cloud, Deployment & MLOps: AWS (S3, EC2) and Databricks/PySpark — applying in current platform project; Docker containerization, MLflow experiment tracking, Linux virtual machines, scheduled execution, health checks, logging, monitoring, and rollback procedures

Visualization & Applications: Streamlit, Tableau, Plotly, D3.js, SAP Analytics, advanced Excel — interactive dashboards, decision-support apps, executive reporting

Enterprise Systems: SAP, ProPricer, NGApprove, Company Props

FEATURED DATA SCIENCE & ENGINEERING PROJECT

Synthetic Aircraft Sustainment Analytics Platform

Independent Project | In Active Development

- Building an end-to-end Python and SQL pipeline that ingests, validates, and stores large synthetic aircraft, flight, component, maintenance, and inventory datasets (CSV, JSON, fixed-width .DAT) in a PostgreSQL star schema with schema validation, rejected-record quarantine, data-quality metrics, logging, lineage, and idempotent loads.
- Migrating raw-data landing and large-scale transformations to AWS (S3) and Databricks (PySpark) to mirror production cloud analytics workflows and support scalable, reproducible processing.
- Developing classification models that flag maintenance events at risk of exceeding 48-hour turnaround, alongside reliability metrics, anomaly detection, and time-series forecasting of component demand to identify failure drivers.
- Deploying insights through interactive Streamlit and Plotly/Tableau decision-support dashboards; containerizing services with Docker for scheduled execution on a Linux virtual machine with health checks, backups, security patching, and rollback, applying MLOps practices including MLflow model versioning.

PROFESSIONAL EXPERIENCE

Program Cost Analyst

Jul 2024 – Present

Northrop Grumman | San Diego, CA

- Work directly with program managers, engineers, finance, and scheduling stakeholders to scope problems, formulate the right analytical questions, and translate program challenges into rigorous, repeatable analyses.
- Integrate SAP actuals, budgets, schedule inputs, funding data, and ETC/EAC forecasts into recurring earned-value-based evaluations of cost, schedule, and program performance.
- Critically evaluate data quality and relevance — cleaning, reconciling, and validating heterogeneous sources while challenging inconsistent assumptions, mapping differences, and timing issues before results inform decisions.
- Develop financial forecast scenarios from historical performance, current execution conditions, and forward-looking assumptions to drive estimate-at-completion and funding decisions.
- Deliver insights through multiple channels — executive-ready reports, visualizations, and direct consultation with program leadership — translating complex program data into concise, actionable recommendations for non-technical stakeholders.
- Own analytical outputs end to end: maintain traceability between source records, transformations, assumptions, and reported results, and stand behind methodology in reviews — supporting reproducibility, governance, and audit readiness.

Contracts Analyst

Jul 2023 – Jul 2024

Northrop Grumman | San Diego, CA

- Translated ambiguous, evolving contract requirements (terms, modifications, deliverables, FAR/DFARS) into structured execution guidance — bridging program, technical, legal, and business audiences.
- Analyzed change impacts to identify dependencies, compliance risks, and downstream effects on program execution, data needs, and reporting.
- Maintained controlled, traceable, audit-ready records supporting governance and reliable decision-making in a regulated DoD environment.

Pricing Analyst

Aug 2022 – Jul 2023

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- Integrated historical actuals, labor estimates, material inputs, rates, and schedules into structured cost models for multiple concurrent DoD proposals; applied cost-driver evaluation, scenario modeling, and sensitivity analysis to build defensible estimates.
- Challenged assumptions in Basis of Estimates — flagging unsupported inputs, inconsistent methodologies, data-quality concerns, and FAR Part 15 compliance risk.
- Communicated model methodology, assumptions, limitations, and findings to stakeholders during reviews, audits, fact-finding, and negotiations.
- Trained and mentored new analysts on analytical tools, data validation, and quality control; led a sister-site audit driving standardization and adoption of best practices.

ADDITIONAL DATA SCIENCE PROJECTS

Steam Recommendation System

UC San Diego

- Built an end-to-end Python recommendation workflow on large-scale user-item interaction data using SVD, SVD++, and Bayesian Personalized Ranking; achieved ~0.864 AUC with the strongest-performing approach.
- Designed repeatable training and evaluation workflows; compared ranking performance, scalability, sparsity, interpretability, and cold-start tradeoffs, documenting assumptions, methodology, and limitations.

NETCLIX Interactive Data Visualization

UC San Diego

- Developed a browser-based interactive visualization application (JavaScript, D3.js, HTML/CSS) enabling filtering, comparison, aggregation, and user-driven exploration of multidimensional data for technical and non-technical users.

EDUCATION

Master of Data Science — University of California San Diego

Mar 2026

Machine learning, statistical analysis, data engineering, data visualization, recommendation systems, databases, applied analytics.

B.S. Business Administration - Information Systems, Honors — San Diego State University

May 2022

Fowler College of Business | Campus GPA: 4.0 | Summa Cum Laude | Systems analysis, database management, programming, business intelligence, application development.

CLEARANCE, CERTIFICATIONS & RECOGNITION

- Active Security Clearance
- Northrop Grumman BRAVO Award — exceptional performance and support of stretch assignments
- Northrop Grumman Pathways Program — completed rotations in pricing, contracts, and program control
- Northrop Grumman ECTAC Certification — contracts execution and compliance practices